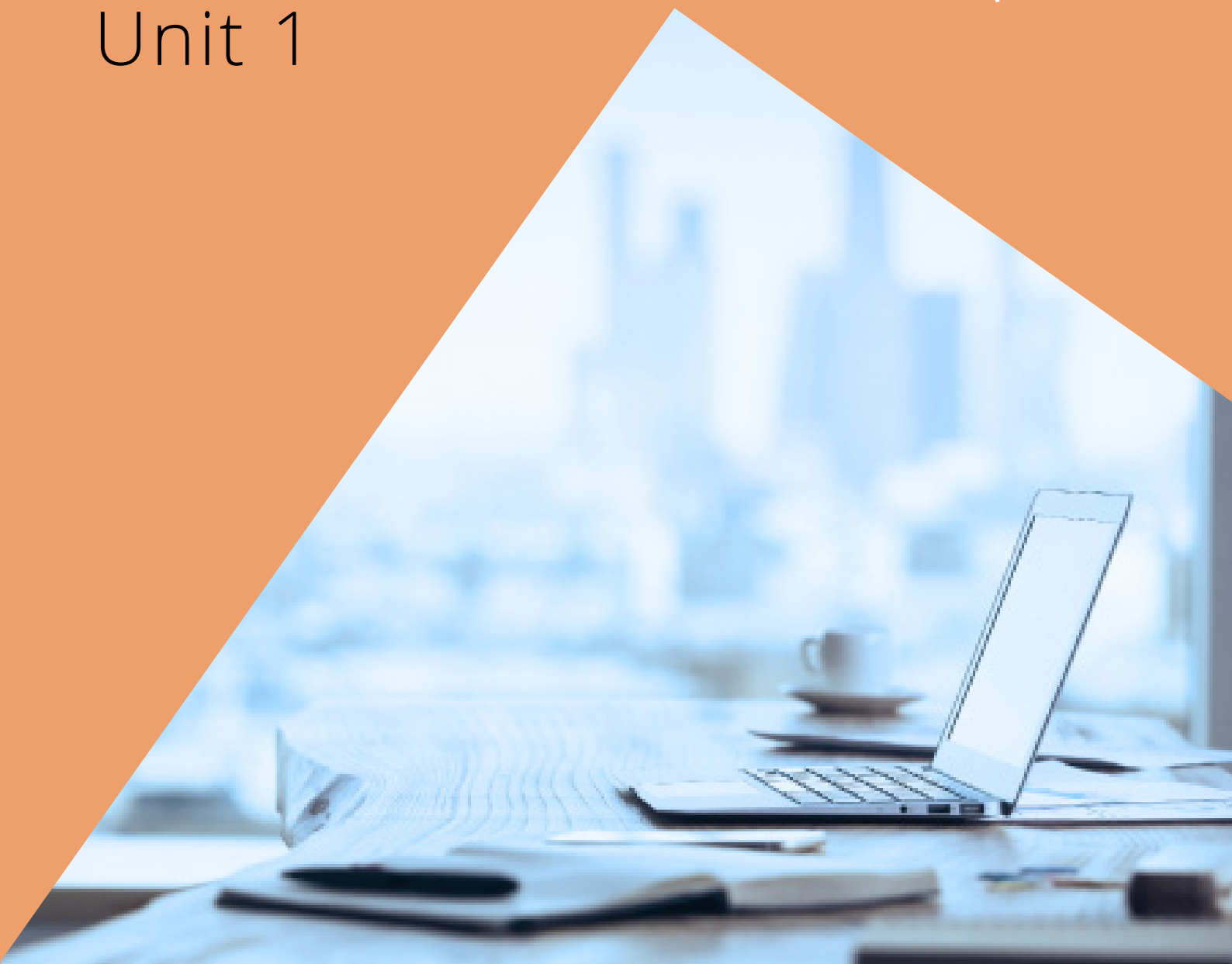


vision**2**learn

Understanding Coding Level 2

Hints
and
Tips

Unit 1



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Hints and Tips Unit 1

The information in this guide will be helpful to you when you are ready to complete your assessment for the following unit:

Unit 1: Understanding the principles of coding
Level 2

Section 1 – Know about coding languages

- Q1.** In the box below state the meaning of the term **coding** and give an example of where and why you would use a **coding language**.

This question covers both learning outcomes 1.1 and 1.2. For this you simply need to define what coding means and what a coding language is. You are asked to give an example of where you would use a coding language so think about what coding languages are used for, what is their purpose?

- Q2.** In the box below, **identify** at least two different **coding languages** that can be used to code.

This is an identify question so you need to simply give the names of different coding languages. You must give two as a minimum but try to provide as many different languages as you can.

- Q3.** Do a little research to find examples of procedural and object-oriented languages. In the box below **state** the differences between the two. (Make sure you evidence websites and/or books used).

As suggested, you will find it useful to carry out research to find examples of procedural and object-oriented languages. You should then consider what is the difference between them? What tasks or functions are each type commonly used for? Try to be as detailed as you can in your response.

- Q4.** Do a little research to find examples of syntax used in coding languages, use this research to help you to **define** what is meant by syntax in the box below.

For this question you need to define what is meant by the term syntax. You may find it helpful to carry out research on this and as well as giving examples of syntax used in coding.

Q5. In the box below, **list** examples of technologies in the home that uses code to run.

For this question think about the different devices that you have in your home that require coding to make them operate. Try to list as many different devices as you can.

Section 2 – Understand job functions in coding

- Q1.** In the box below, from your research, **list** the **key job functions** and the **key skills** that would be required for a coding professional.

This is an identify question so you only need to list your answer. Try to find as many different job functions as you can. Carrying out research will help you with this. Think about all the different key skills that are also required not just coding or programming skills.

- Q2.** In the box below, from your research, **list** the **different roles** that you would find in a **coding team**.

As with question 1, this is an identify question so you can simply list your responses. Try to think of as many different roles as you can. Carrying out research will help you with this.

- Q3.** In the box below, **explain** in your own words why **team working**, and **project management skills** are important in coding.

This is an explain question so you should try and provide a good level of detail as possible in your answer. For team working, think about what working in a team will achieve what benefits does it have? For project management skills, why are these skills important what benefit does it bring?

- Q4.** In the box below, from your research, **list** the **different careers** and **career progression routes**.

This is another identify question so you only need to list your response. Carry out research on the different careers that are available to programmers and the various progression routes that are available to those working in these roles.

Section 3 – Understand the software development lifecycle

- Q1.** In the table below **identify** and **explain** in your own words each of the key stages of the software development lifecycle.

There are two parts to this question. In the first column you simply need to state the steps of the software development cycle. For the second part you then need to explain what each stage is. Think about what happens at each stage? It's important that your explanation is clear, detailed and that it is in your own words.

- Q2.** The **planning stage** covers the requirements gathering, analysis, and design stages. Do a little research on this stage and in the box below using your own words, **describe** why the planning stage is important.

For question 1 you had to identify the different stages but for this question you simply need to focus on the planning stage and why it is so important. Think about what happens at this stage, why is it so useful, what outcomes, both positive and negative are associated with this stage? This is a describe question so try and provide good detail in your response.

- Q3.** In the box below **list** the actions that take place during the **implementation stage**.

This is another identify question so you only need to list your response. You simply need to think about the different actions that are required in the implementation phase.

- Q4.** Do a little research on the testing stage and in the box below using your own words, **explain** why the **testing stage** is important.

This is an explain question so try to provide good detail in your answer. You need to focus only on the testing stage and consider what makes it so important. What does the testing stage achieve? What will it help with? What will it help you to find?

- Q5.** In the box below using your own words, **describe** the actions that would be taken during the **deployment stage**.

This is a describe question so try to provide good detail in your response. Think about the actions that are taken during this stage. What happens during the deployment stage?

- Q6.** Do a little research on the maintenance stage and in the box below using your own words, **explain** why the maintenance stage is important.

This is an explain question so try to provide good detail in your answer. You need to focus only on the maintenance stage and consider what makes it so important. What happens at this stage? Think about the ongoing nature of maintenance. What are the potential consequences if this is not done?